# Math 110 — Week 8

## Generalized eigenvectors and Jordan form (Notes Ch. 14–15, LADR 8A–8C, HW8)

## §1 Facts used from earlier weeks

$V$ is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "$S$ kills $v$" means $Sv = 0$, and "$q$ kills $T$" means $q(T) = 0$.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

**Why.**

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of $V$. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: $T$ of any vector is a combination of the $Tu_i$ (which are 0) and the $Tv_j$.
3. Independent: if $\sum c_j Tv_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the $u_i$, and independence of the basis of $V$ gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

**Why.** $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of $n$ vectors is a basis.

(b) A subspace of dimension $n$ is all of $V$.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

**Why.**

1. (a), (b): an independent list extends to a basis, and every basis has length $n$.
2. (c): a basis of $U$ followed by a basis of $W$ is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length $n$, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis $x_i, u_j$ of $U$ and, separately, to a basis $x_i, w_l$ of $W$. Together the $x_i, u_j, w_l$ span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ ($x, u, w$ combinations of the $x_i$, $u_j$, $w_l$), then $w = -x - u \in U \cap W$, so $w$ is a combination of the $x_i$. The basis $x_i, w_l$ of $W$ is independent, so $w = 0$. Then $x + u = 0$, and independence of $x_i, u_j$ gives $x = u = 0$.

(T4) **Products of operators.** If $S, T$ are injective, so is $ST$. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

**Why.** $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in $T$ commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are $T$-invariant.

**Why.** Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda)=0 \iff (z-\lambda) \mid p$.

**Why.** Divide $p$ by $z-\lambda$: the remainder is a constant, and evaluating at $\lambda$ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) $\lambda$ is an eigenvalue $\iff$ $T-\lambda I$ is not injective.

(b) If $Tv=\lambda v$, then $p(T)v=p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

**Why.**

1. (a) is the definition: $Tv=\lambda v$ with $v\neq 0$ means $(T-\lambda I)v=0$ with $v\neq 0$.
2. (b): $T^k v=\lambda^k v$ by induction; add up the terms of $p$.
3. (d): take $v\neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v=0$ for some $p\neq 0$; $p$ is not constant since $v\neq 0$.
4. Factor $p=c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v\neq 0$ to 0, so some $T-\lambda_j I$ is not injective (T4).

(T8) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1,\ldots,v_n \iff$ every $\operatorname{span}(v_1,\ldots,v_j)$ is invariant. Then the eigenvalues of $T$ are the diagonal entries. Over $\mathbb{C}$ every operator has an upper-triangular matrix.

**Why.**

1. Column $j$ holds the coordinates of $Tv_j$, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1,\ldots,v_j)$ for every $j$.
2. If $a_{jj}=\lambda$: $T-\lambda I$ maps $\operatorname{span}(v_1,\ldots,v_j)$ into $\operatorname{span}(v_1,\ldots,v_{j-1})$, a smaller space, so it is not injective (T1). So $\lambda$ is an eigenvalue.
3. If no $a_{jj}$ equals $\lambda$: $\mathcal{M}(T-\lambda I)$ is triangular with nonzero diagonal. Its columns are independent: in a relation, the right-most column used is the only one with a nonzero entry in its own diagonal row. So $T-\lambda I$ is injective.
4. Existence over $\mathbb{C}$, by induction on $n$: take an eigenvalue $\lambda$ (T7(d)). $U=\operatorname{range}(T-\lambda I)$ is invariant (T5) and $\dim U<n$, since $T-\lambda I$ is not injective (T1).
5. By induction $U$ has a basis $u_1,\ldots,u_m$ with every $\operatorname{span}(u_1,\ldots,u_j)$ invariant. Extend it to a basis $u_1,\ldots,u_m,v_1,\ldots,v_k$ of $V$.
6. $Tv_j=(T-\lambda I)v_j+\lambda v_j \in U+\operatorname{span}(v_j) \subseteq \operatorname{span}(u_1,\ldots,u_m,v_1,\ldots,v_j)$. So every span of an initial part of the list is invariant.

(T9) **Block diagonal matrices.** If $V=U\oplus W$ with both $T$-invariant, then in a basis of $U$ followed by one of $W$, $\mathcal{M}(T)=\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

**Why.** $Tu \in U$ is a combination of the $U$-basis only, so its column is 0 in the $W$-rows (same for $W$). If $T(u+w)=0$, then $Tu=-Tw \in U\cap W=\{0\}$.

(T10) **Change of basis.** If column $j$ of $S$ holds the $B$-coordinates of the $j$-th vector of $B'$, then $\mathcal{M}_{B'}(T)=S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

**Why.** $S$ converts $B'$-coordinates to $B$-coordinates: apply $S$, then $\mathcal{M}_B(T)$, then convert back with $S^{-1}$. Every invertible $S$ arises this way, since its columns form a basis.

(T11) **Minimal polynomial (Week 7).** $m_T$ is the monic polynomial of least degree with $m_T(T)=0$. (a) $q(T)=0 \iff m_T \mid q$ (Prop. 65). (b) The roots of $m_T$ in $\mathbb{F}$ are exactly the eigenvalues (Prop. 66). (c) $\deg m_T \leq n$ (Prop. 67). (d) If $U$ is invariant, $m_{T|_U} \mid m_T$. (e) If $V=U_1\oplus\cdots\oplus U_r$ with every $U_i$ invariant, $m_T$ is the least common multiple of the $m_{T|_{U_i}}$.

**Why.** (a) Divide $q$ by $m_T$; the remainder is 0 at $T$ and has smaller degree, so it is 0. (d) $m_T(T|_U)u = m_T(T)u = 0$, then (a). (e) $q(T) = 0$ exactly when $q(T|_{U_i}) = 0$ for all $i$, i.e. when every $m_{T|_{U_i}}$ divides $q$. (b), (c) are proved in the Week 7 handout.

(T12) **Direct sums of several subspaces (Week 1).** $U_1+\cdots+U_m$ is direct if the only way to write $0 = u_1+\cdots+u_m$ with $u_k \in U_k$ is with every $u_k = 0$ (Prop. 7, (iii)⇒(i)). If $V = U_1 \oplus \cdots \oplus U_m$, bases of the $U_k$ put together form a basis of $V$.

**Why.** Def. 6 asks $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for each $k$. If $v$ is in that intersection, write $v = u_1 + \cdots + u_{k-1}$; then $0 = u_1 + \cdots + u_{k-1} + (-v)$ with $-v \in U_k$, so every term is 0, and $v = 0$. Bases: the combined list spans, since every $v = u_1 + \cdots + u_m$ and each $u_k$ is a combination of its basis. It is independent: group a relation by subspace as $w_1 + \cdots + w_m = 0$ with $w_k \in U_k$; directness gives every $w_k = 0$ (Prop. 7), and each $w_k = 0$ is a relation in a basis of $U_k$, so its coefficients are 0.

# §2 This week

## §2.1 Generalized eigenvectors (Ch. 14)

**The question.** For $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ every eigenvector is a multiple of $e_1$, so $A$ has no eigenbasis. But $(A-2I)e_2 = e_1$ and $(A-2I)^2 e_2 = 0$. If we also accept vectors $v$ with $(A-\lambda I)^k v = 0$ for some $k \geq 1$, does every complex operator have a basis of such vectors, for which $\lambda$ do they exist, and how many independent ones are there for each $\lambda$?

**Definition** (Def. 37 / LADR 8.8, 8.19)**.** $v \neq 0$ is a **generalized eigenvector** for $\lambda$ if $(T-\lambda I)^k v = 0$ for some $k \geq 1$. The **generalized eigenspace** is $G(\lambda, T) = \bigcup_{k \geq 0} \operatorname{null}(T-\lambda I)^k$.

- $E(\lambda, T) \subseteq G(\lambda, T)$ (take $k = 1$). For $N = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$, $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$, since $N^2 = 0$.
- A union over all $k$ is awkward. Props. 69–70 show it stops at a fixed power.
- An operator $N$ is **nilpotent** if $N^k = 0$ for some $k$ (Def. 39), for example $N$ above. So $G(\lambda, T)$ consists of the vectors sent to 0 by some power of $T - \lambda I$, and $T - \lambda I$ restricted to it is nilpotent (Prop. 70).

**Proposition** (Prop. 69 / LADR 8.1, 8.2: the null-space chain)
$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T \subseteq \operatorname{null} T^2 \subseteq \cdots$. If $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^m = \operatorname{null} T^{m+k}$ for all $k \geq 0$.

**The idea**

A vector killed by $T^k$ stays killed by $T^{k+1} = T \cdot T^k$.

Only one equality, $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. To use it on a vector $v \in \operatorname{null} T^{m+k+1}$, we need some vector in $\operatorname{null} T^{m+1}$ built from $v$.

Split the power: $T^{m+k+1}v = T^{m+1}(T^k v)$. So $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$, which says $T^{m+k}v = 0$. That is the equality one step higher; repeat.

For example, the shift on $\mathbb{F}^3$ ($e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$) has $\dim \operatorname{null} T^k = 0, 1, 2, 3, 3, 3, \ldots$: it grows until two neighbours are equal, and after that all are equal.

*Proof.* 1. If $T^k v = 0$, then $T^{k+1}v = T(0) = 0$. So $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$.

2. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $T^{m+k+1}v = 0$. Then $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$.

3. So $T^{m+k}v = T^m(T^k v) = 0$. Hence $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, and with step 1 they are equal. So if $\operatorname{null} T^{m+k} = \operatorname{null} T^m$, then $\operatorname{null} T^{m+k+1} = \operatorname{null} T^m$; by induction on $k$, $\operatorname{null} T^{m+k} = \operatorname{null} T^m$ for all $k$.

□

**Proposition** (Prop. 70 / LADR 8.3, 8.20)
$\operatorname{null} T^n = \operatorname{null} T^{n+1}$. Hence $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, a $T$-invariant subspace.

**The idea**

Until it stops, each step raises the dimension by at least 1 (a strictly larger subspace has larger dimension). Dimensions are at most $n$. Once it stops, it stays stopped (Prop. 69).

Count: starting from dimension 0, $n+1$ strict steps would reach dimension $n+1 > n$. So one of the first $n+1$ steps is an equality, and from there on the chain is constant.

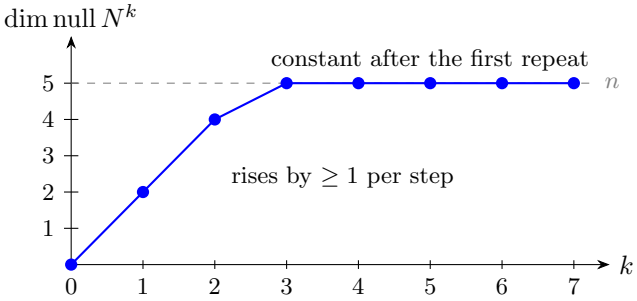
So the union defining $G(\lambda, T)$ is then the single null space $\operatorname{null}(T - \lambda I)^n$. So it is a subspace, and it is invariant by (T5).

*Proof.* 1. Suppose $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$. By Prop. 69 no earlier pair is equal either.

2. Then
$$0 = \dim \operatorname{null} T^0 < \dim \operatorname{null} T^1 < \cdots < \dim \operatorname{null} T^{n+1} \leq n :$$
that is $n+2$ distinct integers in $\{0, \ldots, n\}$. Impossible.

3. Apply this to $T - \lambda I$. For $k \leq n$, $\operatorname{null}(T - \lambda I)^k \subseteq \operatorname{null}(T - \lambda I)^n$ (Prop. 69), and for $k \geq n$, $\operatorname{null}(T - \lambda I)^k = \operatorname{null}(T - \lambda I)^n$ (Prop. 69 with $m = n$). So $G(\lambda, T) = \bigcup_k \operatorname{null}(T - \lambda I)^k = \operatorname{null}(T - \lambda I)^n$. It is invariant by (T5).

□




$N$ on $\mathbb{F}^5$ with $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$, $Ne_4 = 0$, $Ne_5 = e_4$. Then $\operatorname{null} N = \operatorname{span}(e_1, e_4)$, $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2, e_4, e_5)$, $\operatorname{null} N^3 = \mathbb{F}^5$, so $\dim \operatorname{null} N^k = 0, 2, 4, 5, 5, \ldots$ While the chain grows it gains at least 1 per step, and it cannot pass $n$, so it has stopped by $k = n$ (Prop. 70). Once two neighbours agree, all later ones agree (Prop. 69).

**Example 61: computing $G(1, A)$.** Take
$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$
and $\lambda = 1$.

1. The powers of $A - I$ are
$$A - I = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad (A - I)^2 = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 4 \end{pmatrix}, \quad (A - I)^3 = \begin{pmatrix} 0 & 0 & 6 \\ 0 & 0 & 4 \\ 0 & 0 & 8 \end{pmatrix}.$$

2. Their null spaces are $\operatorname{span}(e_1)$, then $\operatorname{span}(e_1, e_2)$, then $\operatorname{span}(e_1, e_2)$ again.

3. Two consecutive null spaces agree, so the chain has stopped (Prop. 69): $G(1, A) = \operatorname{span}(e_1, e_2)$.

**Proposition** (LADR 8.4, 8.5: splitting off the nilpotent part)
(a) $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$. (b) $T$ is injective on $\operatorname{range} T^n$.

**The idea**

The aim is to split $V$ into a part where $T$ is nilpotent ($\operatorname{null} T^n$) and a part where $T$ is injective ($\operatorname{range} T^n$). Applied to $T - \lambda I$, this splits off $G(\lambda, T)$.

$\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1). So by (T3) it is enough to show $\operatorname{null} T^n \cap \operatorname{range} T^n = \{0\}$.

A first try is the power 1: is $\operatorname{null} T \cap \operatorname{range} T = \{0\}$? No. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_1 = Ne_2$ is in the range and $Ne_1 = 0$. A nonzero vector of range $N$ can lie in null $N$.

With the power $n$, $\operatorname{range} T^n$ meets $\operatorname{null} T^n$ only in 0, because the chain has stopped. If $v = T^n u$ and $T^n v = 0$, then $T^{2n} u = 0$, and $\operatorname{null} T^{2n} = \operatorname{null} T^n$ (Prop. 70) gives $T^n u = 0$, i.e. $v = 0$.

*Proof.* 1. Let $v = T^n u$ with $T^n v = 0$. Then $T^{2n} u = 0$, so $u \in \operatorname{null} T^{2n} = \operatorname{null} T^n$, so $v = 0$.

2. (a) $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1), so $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (T3).

3. (b) If $w \in \operatorname{range} T^n$ and $Tw = 0$, then $w \in \operatorname{null} T \subseteq \operatorname{null} T^n$, so $w = 0$.

□

**Proposition** (Prop. 71)
$G(\lambda, T) \neq \{0\} \iff \lambda$ is an eigenvalue of $T$.

**The idea**

Apply $T - \lambda I$ to $v$ repeatedly: $v \mapsto (T - \lambda I)v \mapsto \cdots \mapsto 0$. The last nonzero vector in this list is sent to 0 by $T - \lambda I$, so it is an eigenvector.

*Proof.* 1. **Assume $\lambda$ is an eigenvalue.** Then $\{0\} \neq E(\lambda, T) \subseteq G(\lambda, T)$.

2. **Assume $G(\lambda, T) \neq \{0\}$.** Take $v \neq 0$ in it. Some power of $T - \lambda I$ sends $v$ to 0, and the power 0 does not, so there is a largest $j \geq 0$ with $w = (T - \lambda I)^j v \neq 0$.

3. By the choice of $j$, $(T - \lambda I)w = (T - \lambda I)^{j+1} v = 0$ and $w \neq 0$, so $w$ is an eigenvector for $\lambda$.

□

$$v \xrightarrow{T-\lambda I} (T-\lambda I)v \xrightarrow{T-\lambda I} \cdots \xrightarrow{T-\lambda I} w \xrightarrow{T-\lambda I} 0$$

generalized eigenvector — last nonzero vector: an eigenvector

Repeatedly applying $T - \lambda I$ to a generalized eigenvector reaches 0. The last nonzero vector $w$ has $(T - \lambda I)w = 0$, so it is an eigenvector (Prop. 71). Smallest case: $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_2 \mapsto e_1 \mapsto 0$.

**Proposition** (Prop. 72 / LADR 8.12)

Generalized eigenvectors $v_1, \ldots, v_m$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ are independent.

**The idea**

How independence of eigenvectors is proved (Prop. 54): apply an operator that kills every term except the $k$-th and is nonzero on the $k$-th. What is left is $a_k \cdot (\text{nonzero vector}) = 0$.

A first try: $(T - \lambda_i I)^n$ kills $v_i$, so $S = \prod_{i \neq k} (T - \lambda_i I)^n$ kills every $v_i$ with $i \neq k$. But it is not clear what $S$ does to $v_k$, which is only a generalized eigenvector.

On an actual eigenvector $w$ for $\lambda_k$, $S$ is easy:

$$Sw = \prod_{i \neq k} (\lambda_k - \lambda_i)^n w \neq 0$$

(T7). So first turn $v_k$ into an eigenvector by applying $(T - \lambda_k I)^N$, as in Prop. 71. The extra factor still gives $Sv_i = 0$ for $i \neq k$, because all the factors commute.

*Proof.* 1. Fix $k$; we show $a_k = 0$. Let $N$ be the largest exponent with $w = (T - \lambda_k I)^N v_k \neq 0$ (it exists: $v_k \neq 0$, and some power of $T - \lambda_k I$ sends $v_k$ to 0). By the choice of $N$, $(T - \lambda_k I)w = 0$, so $w$ is an eigenvector: $Tw = \lambda_k w$.

2. On an eigenvector, each $T - \lambda_i I$ multiplies by the number $\lambda_k - \lambda_i$ (T7). So $(T - \lambda_i I)^n w = (\lambda_k - \lambda_i)^n w$, and this number is not 0 for $i \neq k$.

3. Let
$$S = \prod_{i \neq k} (T - \lambda_i I)^n \cdot (T - \lambda_k I)^N$$
and apply $S$ to $\sum_i a_i v_i = 0$.

4. *Terms with $i \neq k$.* The factors of $S$ commute (T5), so let $(T - \lambda_i I)^n$ act first on $v_i$. It gives 0, because $v_i \in G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). So $Sv_i = 0$.

5. *The term with $i = k$.* $(T - \lambda_k I)^N v_k = w$, and then by step 2,
$$Sv_k = \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w.$$

6. So
$$0 = a_k \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w.$$
The product is a nonzero number and $w \neq 0$, so $a_k = 0$.

□

**Two consequences.**

- $G(\lambda, T) \cap G(\mu, T) = \{0\}$ for $\lambda \neq \mu$: a nonzero $v$ in both would make the list $v, v$ independent (Prop. 72). So a generalized eigenvector belongs to only one eigenvalue.
- If $U$ is invariant, $G(\lambda, T|_U) = G(\lambda, T) \cap U$, since $(T|_U - \lambda I)^k u = (T - \lambda I)^k u$ for $u \in U$.

**Theorem** (Prop. 73 / LADR 8.9, 8.22: generalized eigenspace decomposition)
Over $\mathbb{C}$, if $\lambda_1, \dots, \lambda_m$ are the distinct eigenvalues of $T$, then $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. In particular $V$ has a basis of generalized eigenvectors of $T$.

**The idea**

Two things are needed: the $G(\lambda_i, T)$ add up to $V$, and the sum is direct.

Splitting off the nilpotent part, (a), applied to $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$ with $U = \operatorname{range}(T - \lambda_1 I)^n$ invariant. $U$ is smaller than $V$ because $G(\lambda_1, T) \neq 0$.

Spanning: induct on $\dim V$. The theorem for $T|_U$ writes $U$ as a sum of generalized eigenspaces of $T|_U$, and these lie inside those of $T$. The eigenvalue $\lambda_1$ does not come back in $U$, because $T - \lambda_1 I$ is injective on $U$. So every $v = g + u$ with $g \in G(\lambda_1, T)$, $u \in U$ is a sum of vectors from the $G(\lambda_i, T)$.

Direct: by (T12) it is enough that $0 = u_1 + \cdots + u_m$ with $u_i \in G(\lambda_i, T)$ forces every $u_i = 0$. If some $u_i$ were nonzero, the nonzero ones would be generalized eigenvectors for distinct eigenvalues whose sum is 0, a dependence that Prop. 72 rules out.

**Why $\mathbb{C}$.** We need an eigenvalue $\lambda_1$ to start (T7). Over $\mathbb{R}$, the rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has no eigenvalues, so all its generalized eigenspaces are 0.

**Example: the decomposition for the matrix of Ex. 61.** For $A$ of Ex. 61 (above), $G(1, A) = \operatorname{span}(e_1, e_2)$, and $G(3, A) = E(3, A) = \operatorname{span}\big((3, 2, 4)\big)$. So $\mathbb{C}^3 = G(1, A) \oplus G(3, A)$.

*Proof.* 1. Induction on $n$; $n = 1$ is clear. Take an eigenvalue $\lambda_1$ (T7).

2. Splitting off the nilpotent part, (a), applied to $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$, $U = \operatorname{range}(T - \lambda_1 I)^n$. $U$ is invariant (T5), and $\dim U < n$ since $G(\lambda_1, T) \neq 0$ (Prop. 71).

3. *Which eigenvalues does $T|_U$ have?* An eigenvector of $T|_U$ is an eigenvector of $T$, so they are among $\lambda_1, \dots, \lambda_m$. And $\lambda_1$ is not one of them: $T - \lambda_1 I$ is injective on $U = \operatorname{range}(T - \lambda_1 I)^n$ (splitting off the nilpotent part, (b), applied to $T - \lambda_1 I$).

4. *$U$ is a sum of generalized eigenspaces of $T$.* $U$ is invariant, so $T|_U$ is an operator on the complex space $U$, and $\dim U < n$. By the induction hypothesis, $U$ is the sum of the generalized eigenspaces of $T|_U$ for its eigenvalues, which are among $\lambda_2, \dots, \lambda_m$ by step 3. Adding $G(\lambda_i, T|_U) = \{0\}$ for each $\lambda_i$ that is not an eigenvalue of $T|_U$ (Prop. 71) changes nothing, so
$$U = G(\lambda_2, T|_U) + \cdots + G(\lambda_m, T|_U).$$
Each $G(\lambda_i, T|_U)$ is contained in $G(\lambda_i, T)$, because $(T|_U - \lambda_i I)^k u = (T - \lambda_i I)^k u$ for $u \in U$.

5. *The $G(\lambda_i, T)$ add up to $V$.* Let $v \in V$. By step 2, $v = g + u$ with $g \in G(\lambda_1, T)$ and $u \in U$. By step 4, $u = u_2 + \cdots + u_m$ with $u_i \in G(\lambda_i, T|_U) \subseteq G(\lambda_i, T)$. So
$$v = g + u_2 + \cdots + u_m \in G(\lambda_1, T) + \cdots + G(\lambda_m, T).$$
So $V \subseteq G(\lambda_1, T) + \cdots + G(\lambda_m, T)$. Conversely, each $G(\lambda_i, T)$ is a subspace of $V$ and $V$ is closed under addition, so $G(\lambda_1, T) + \cdots + G(\lambda_m, T) \subseteq V$. Hence $V = G(\lambda_1, T) + \cdots + G(\lambda_m, T)$.

6. *The sum is direct.* By (T12) it is enough to show: if $0 = u_1 + \cdots + u_m$ with $u_i \in G(\lambda_i, T)$, then every $u_i = 0$. Suppose not, and let $u_{i_1}, \dots, u_{i_k}$ ($k \geq 1$) be the nonzero ones. They are generalized eigenvectors for the distinct eigenvalues $\lambda_{i_1}, \dots, \lambda_{i_k}$, so they are independent (Prop. 72). But
$$1 \cdot u_{i_1} + \cdots + 1 \cdot u_{i_k} = 0$$
is a relation with nonzero coefficients, a contradiction. So every $u_i = 0$, and $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$.

□

On each $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with $N_i$ nilpotent. So everything reduces to nilpotent operators (Ch. 15).

**Definition** (Def. 38)**. Algebraic multiplicity** of $\lambda$: $\dim G(\lambda, T)$. **Geometric multiplicity**: $\dim E(\lambda, T)$.

For the question's $A$: $G(2, A) = \mathbb{C}^2$, so the algebraic multiplicity of 2 is 2 and the geometric multiplicity is 1.

**The question answered.** Yes over $\mathbb{C}$: $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73). Such vectors exist exactly for the eigenvalues (Prop. 71), and for $\lambda$ there are $\dim G(\lambda, T)$ independent ones, the algebraic multiplicity.

- geometric $\leq$ algebraic, since $E \subseteq G$.
- Over $\mathbb{C}$, the algebraic multiplicities add up to $n$ (Prop. 73).
- Over $\mathbb{C}$, $T$ is diagonalizable $\iff$ geometric = algebraic for every eigenvalue.

  **Why.** The sum $\sum E(\lambda_i, T)$ is direct (T7), and $T$ is diagonalizable iff it is $V$, i.e. $\sum \dim E = n = \sum \dim G$. Since each $\dim E \leq \dim G$, the sums agree iff every term agrees.

**Theorem** (Prop. 74 / LADR 8.26, 8.29: Cayley–Hamilton)
Over $\mathbb{C}$, let $\mu_i = \dim G(\lambda_i, T)$ and $\chi_T(x) = \prod_i (x - \lambda_i)^{\mu_i}$ (the **characteristic polynomial**). Then $\chi_T(T) = 0$, so $m_T \mid \chi_T$. Also $\deg \chi_T = \sum \mu_i = n$ (Prop. 73), so $\deg m_T \leq n$.

**The idea**

$V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73), so it is enough to take $v$ in one $G(\lambda_i, T)$. There, $T - \lambda_i I$ is nilpotent.

Which power of $T - \lambda_i I$ kills $G(\lambda_i, T)$? Prop. 70, applied inside $G(\lambda_i, T)$, a space of dimension $\mu_i$, gives the power $\mu_i$. That is why the exponents in $\chi_T$ are the $\mu_i$.

The factors of $\chi_T(T)$ commute, so for $v \in G(\lambda_i, T)$ let $(T - \lambda_i I)^{\mu_i}$ act first.

*Proof.* 1. *Which power of $T - \lambda_i I$ is 0 on $G(\lambda_i, T)$?* Let $N = (T - \lambda_i I)|_{G(\lambda_i, T)}$, an operator on a space of dimension $\mu_i$. Every vector of that space is sent to 0 by some power of $N$. Prop. 70, applied to $N$ on this $\mu_i$-dimensional space, says the power $\mu_i$ is enough: $N^{\mu_i} = 0$.

2. For $v \in G(\lambda_i, T)$, the factors commute (T5), so the factor $(T - \lambda_i I)^{\mu_i}$ can act first, and it sends $v$ to 0:

$$\chi_T(T)v = \prod_{j \neq i} (T - \lambda_j I)^{\mu_j} (T - \lambda_i I)^{\mu_i} v = 0.$$

3. Every $v \in V$ is a sum of vectors from the $G(\lambda_i, T)$ (Prop. 73), and $\chi_T(T)$ is linear and 0 on each of them, so $\chi_T(T)v = 0$: $\chi_T(T) = 0$; then $m_T \mid \chi_T$ (T11(a)) and $\deg \chi_T = \sum \mu_i = n$.

□

- For a matrix, $\chi_A(x) = \det(xI - A)$ (cited; determinants are Ch. 25).
- $2 \times 2$: $\chi_A = x^2 - (\operatorname{tr} A)x + \det A$.

  **Why.** Multiplying out gives $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq cI$, $m_A$ has degree 2 and divides both this polynomial and $\chi_A$, so all three agree. If $A = cI$, both are $(x - c)^2$.
- $\chi_T$ and $m_T$ have the same roots; the exponents in $m_T$ can be smaller.

**Proposition** (LADR 8.31: diagonal of a triangular matrix)
Over $\mathbb{C}$, if $\mathcal{M}(T)$ is upper-triangular, then $\dim G(\lambda, T)$ is the number of times $\lambda$ appears on the diagonal.

**The idea**

Let $c_\lambda$ be the number of times $\lambda$ appears on the diagonal. By Prop. 70 and (T1),

$$\dim G(\lambda, T) = \dim \operatorname{null}(T - \lambda I)^n = n - \operatorname{rank}(T - \lambda I)^n.$$

The matrix of $(T - \lambda I)^n$ is upper-triangular with diagonal entries $(a_{jj} - \lambda)^n$, nonzero in exactly $n - c_\lambda$ places.

Computing the rank exactly is hard: columns with a 0 on the diagonal can still be nonzero and add to the rank.

Settle for an inequality. The $n - c_\lambda$ columns with nonzero diagonal entries are independent (T8), so rank $\geq n - c_\lambda$, i.e. $\dim G(\lambda, T) \leq c_\lambda$. Now add over all $\lambda$: both sides add up to $n$, so none of the inequalities can be strict.

The same argument (prove $\leq$ for each term, then compare totals) also proves the criterion after Def. 38: $T$ is diagonalizable if and only if each geometric multiplicity equals the algebraic multiplicity.

*Proof.* 1. Let $c_\lambda$ be the count. $\mathcal{M}((T - \lambda I)^n)$ is triangular with diagonal $(a_{jj} - \lambda)^n$, nonzero in $n - c_\lambda$ places.

2. Those columns are independent (T8), so rank $\geq n - c_\lambda$ and $\dim G(\lambda, T) \leq c_\lambda$ (Prop. 70, T1).

3. By Prop. 73,
$$n = \sum_\lambda \dim G(\lambda, T) \leq \sum_\lambda c_\lambda = n,$$
so the sums are equal. Since each term satisfies $\dim G(\lambda, T) \leq c_\lambda$, a single strict inequality would make the left sum smaller than $n$. So $\dim G(\lambda, T) = c_\lambda$ for every $\lambda$.

□

**Proposition** (Exponents of $m_T$ over $\mathbb{C}$, M6)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and $\mu_i = \dim G(\lambda_i, T)$. Then $m_T = \prod_i (x - \lambda_i)^{s_i}$, where

$$s_i = \text{the least } k \text{ with } \operatorname{null}(T - \lambda_i I)^k = G(\lambda_i, T), \qquad 1 \leq s_i \leq \mu_i.$$

In particular $m_T \mid \chi_T$, and $m_T$ has the same roots as $\chi_T$.

**The idea**

$V$ splits into the invariant pieces $G(\lambda_i, T)$ (Prop. 73), and the minimal polynomial of a direct sum is the lcm of the pieces' minimal polynomials (T11(e)).

So compute $m$ on each piece separately. On $G(\lambda_i, T)$ the operator $T - \lambda_i I$ is nilpotent, so the minimal polynomial there is a power of $x - \lambda_i$, and the right power is the first one that is 0 on the piece.

*Proof.* 1. $(T - \lambda_i I)^n = 0$ on $G(\lambda_i, T)$ (Prop. 70), so $m_{T|_{G(\lambda_i, T)}}$ divides $(x - \lambda_i)^n$. Its roots are roots of $(x - \lambda_i)^n$, so it is $(x - \lambda_i)^{s_i}$ for some $s_i$ (over $\mathbb{C}$ it splits into linear factors, T6).

2. $s_i$ is the least $k$ with $(T - \lambda_i I)^k = 0$ on $G(\lambda_i, T)$, i.e. with $\operatorname{null}(T - \lambda_i I)^k \supseteq G(\lambda_i, T)$. Since $\operatorname{null}(T - \lambda_i I)^k \subseteq G(\lambda_i, T)$ always, this means equality.

3. By Prop. 73 and T11(e), $m_T$ is the least common multiple of the $(x - \lambda_i)^{s_i}$. The $\lambda_i$ are distinct, so by the formula in the lcm divides every common multiple (Week 7: the exponent of each $x - \lambda$ is the largest exponent among the polynomials) the least common multiple is their product:
$$m_T = \operatorname{lcm}_i (x - \lambda_i)^{s_i} = \prod_i (x - \lambda_i)^{s_i}.$$

4. $s_i \geq 1$ since $G(\lambda_i, T) \neq 0$, and $s_i \leq \mu_i$ by Prop. 70 applied inside the $\mu_i$-dimensional $G(\lambda_i, T)$.

□

**Computing $m_T$ from null-space dimensions.** For each eigenvalue $\lambda$, compute

$$d_k = \dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$$

for $k = 1, 2, \ldots$. The exponent of $x - \lambda$ is the least $k$ with $d_k = d_{k+1}$; equivalently, the least $k$ with $d_k$ equal to the algebraic multiplicity.

**Exponents for the matrix of Ex. 61.** Take $A$ of Ex. 61 again. For $\lambda = 1$: $d_1 = 1$ and $d_2 = 2$, which is the number of 1s on the diagonal, so the exponent is 2. $\lambda = 3$ appears once, so its exponent is 1. So $m_A = (x-1)^2(x-3)$.

- With the multiplicities $\mu_i$ known, Method 2 of Week 7 (eigenvalues, then test candidates for $m_T$) needs only the candidates with $k_i \le \mu_i$, and $\chi_T$ itself never needs testing (Cayley–Hamilton). For the Week 7 matrix

$$\begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

  1 has multiplicity 1, so $(x-3)(x-1)^2$ is never a candidate, and $(x-3)^2(x-1) = \chi_A$ needs no check.
- For $3 \times 3$ $A$ (cited; determinants are Ch. 25): $\chi_A = \det(xI - A) = x^3 - (\operatorname{tr} A)x^2 + c_2 x - \det A$, where $c_2$ is the sum of the determinants of the three $2 \times 2$ matrices left after deleting row $i$ and column $i$, for $i = 1, 2, 3$. This gives the eigenvalues and multiplicities for Method 2 when $A$ is not triangular.

**Shortcuts**

- **Exponent 1 test.** The exponent of $\lambda$ in $m$ is 1 $\iff$ $\dim E(\lambda)$ equals the algebraic multiplicity

$$\iff \operatorname{rank}(A - \lambda I) = n - \mu_\lambda.$$

  One rank computation per eigenvalue settles diagonalizability. Instance:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \operatorname{rank}(A - 2I) = 2 \neq 3 - 2,$$

  so $x - 2$ has exponent 2 and $A$ is not diagonalizable.
- **Never test $\chi$ itself** when computing $m$: it always works (Cayley–Hamilton). If every smaller candidate fails, $m = \chi$. Instance: for the Week 7 matrix above, $(x-3)(x-1)$ fails, so $m = \chi = (x-3)^2(x-1)$ with no further multiplication.
- **Smaller power.** $G(\lambda, T) = \operatorname{null}(T - \lambda I)^\mu$, $\mu$ the algebraic multiplicity, and even $\operatorname{null}(T - \lambda I)^k$ with $k$ the exponent of $x - \lambda$ in $m_T$ (exponents of $m_T$, above). Why: $T - \lambda I$ is nilpotent on the $\mu$-dimensional space $G(\lambda, T)$, so Prop. 70 inside it gives the power $\mu$. Instance: in Ex. 61, $G(1, A) = \operatorname{null}(A - I)^2$; there is no need to compute $(A - I)^3$.
- **Know the target dimension first.** For a triangular matrix, $\dim G(\lambda)$ is the number of $\lambda$'s on the diagonal (LADR 8.31); in general it is the multiplicity of $\lambda$ as a root of $\chi_A = \det(xI - A)$. Stop the chain $\dim \operatorname{null}(A - \lambda I)^k$ as soon as it reaches that number. Instance: in Ex. 61, 1 appears twice on the diagonal, so stop once $\dim \operatorname{null}(A - I)^k = 2$.
- **Last one by subtraction.** The algebraic multiplicities add up to $n$, so the last one is $n$ minus the others. And if the $\dim G(\lambda_i)$ you found already add up to $n$, there are no other eigenvalues. Instance: in Ex. 61, $\dim G(1, A) = 2$ and $n = 3$, so $\dim G(3, A) = 1$.
- **Multiplicity 1.** If the algebraic multiplicity is 1, then $G(\lambda) = E(\lambda)$ (since $1 \le \dim E \le \dim G = 1$). Instance: $G(3, A) = E(3, A) = \operatorname{span}((3, 2, 4))$ in Ex. 61.
- **One eigenvalue.** Over $\mathbb{C}$, if $\lambda$ is the only eigenvalue, $G(\lambda, T) = V$ and $(T - \lambda I)^n = 0$ with no computation. Instance: for $A$ of Ex. 62 (Section 2.2), $(A - I)^3 = 0$.
- **Geometric multiplicity by rank.** $\dim E(\lambda) = n - \operatorname{rank}(A - \lambda I)$, and in the same way $\dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$ (T1). Instance: for $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$, $\operatorname{rank}(A - 2I) = 1$, so $\dim E(2, A) = 1$.

- $G(\lambda)$ **as a range.** Over $\mathbb{C}$,
$$G(\lambda, T) = \operatorname{range} \prod_{\mu \neq \lambda} (T - \mu I)^{m_\mu},$$
and $\bigoplus_{\mu \neq \lambda} G(\mu, T)$ is the null space of that product. Useful when the other eigenvalues are simple: for the $A$ of Ex. 61, $G(1, A) = \operatorname{range}(A - 3I) = \operatorname{span}(e_1, e_2)$. Why, with $P$ the product and $H = \bigoplus_{\mu \neq \lambda} G(\mu, T)$:
  1. $P = 0$ on each $G(\mu, T)$, $\mu \neq \lambda$: let $(T - \mu I)^{m_\mu}$ act first (Prop. 74, step 1).
  2. $P$ is injective on $G(\lambda, T)$: each $T - \mu I$ is, since a nonzero $v \in G(\lambda, T)$ with $Tv = \mu v$ would lie in $G(\lambda, T) \cap G(\mu, T) = 0$.
  3. $G(\lambda, T)$ is invariant (T5), so $P$ maps it injectively into itself, hence onto itself (T2).
  4. $V = G(\lambda, T) \oplus H$ (Prop. 73) and $P = 0$ on $H$, so $\operatorname{range} P = G(\lambda, T)$ and $\operatorname{null} P = H$.
- **Commuting operators.** If $ST = TS$, then $S$ maps $G(\lambda, T)$ into itself (this is Problem G2): $(T - \lambda I)^n Sv = S(T - \lambda I)^n v = 0$. Instance: every polynomial $p(T)$ commutes with $T$, so $p(T)$ maps each $G(\lambda, T)$ into itself.

**Traps**

- $G(\lambda, T)$ can be strictly bigger than $E(\lambda, T)$: $N = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$. Over $\mathbb{C}$ they agree for all $\lambda$ exactly when $T$ is diagonalizable.
- The exponents in $m_T$ are at most the algebraic multiplicities, and they are not the geometric multiplicities: on $\mathbb{F}^4$ with $Ne_2 = e_1$ and $Ne_1 = Ne_3 = Ne_4 = 0$, $m_N = x^2$ while $\dim E(0) = 3$ and $\dim G(0) = 4$.
- A sum of generalized eigenvectors for different eigenvalues is not a generalized eigenvector: for $T = \operatorname{diag}(1, 2)$, $e_1 + e_2$ lies in neither $G(1, T) = \operatorname{span}(e_1)$ nor $G(2, T) = \operatorname{span}(e_2)$.
- Prop. 73 needs eigenvalues: true over $\mathbb{C}$; over $\mathbb{R}$ only when $m_T$ splits. The rotation $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ of $\mathbb{R}^2$ has $m_R = x^2 + 1$ and $G(\lambda, R) = 0$ for every real $\lambda$, so the sum of the generalized eigenspaces is $0 \neq \mathbb{R}^2$. (If $m_T$ splits, every $T|_U$ with $U \neq 0$ invariant has an eigenvalue, because $m_{T|_U} \mid m_T$ (T11(d)) and its roots are eigenvalues (T11(b)); so the proof of Prop. 73 still works.)
- $V = \operatorname{null} T \oplus \operatorname{range} T$ can fail: for $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$. The splitting needs the power $n$.
- $G(\lambda, T)$ is not $\operatorname{null}(T - \lambda I)^g$ with $g$ the geometric multiplicity: for the shift $J$ on $\mathbb{F}^3$ ($Je_1 = 0$, $Je_2 = e_1$, $Je_3 = e_2$), $g = 1$, but $\operatorname{null} J = \operatorname{span}(e_1) \neq G(0, J) = \mathbb{F}^3$.

## §2.2 Nilpotent operators and Jordan form (Ch. 15)

**The question.** $I$, $B$ and $C$ below all have 1 as their only eigenvalue, and $e_1, e_2, e_3$ are generalized eigenvectors of all three, so a basis of generalized eigenvectors does not tell them apart:

$$B = \begin{pmatrix} 3 & -1 & -1 \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 3 & -2 & 0 \\ 1 & 1 & -1 \\ 0 & 2 & -1 \end{pmatrix}.$$

Is there a basis in which each has a matrix so simple that two matrices are similar exactly when these matrices agree, and what is it for $B$ and for $C$?

**Definition** (Def. 39, continued)**.** For a nilpotent $N$ (Section 2.1), the least $k$ with $N^k = 0$ is the **nilpotency index**.

- $N^n = 0$: $V = G(0, N) = \operatorname{null} N^n$ (Prop. 70). So $m_N = x^k$, $k \leq n$.
- The only eigenvalue is 0 (if $V \neq 0$): $Nv = \lambda v \Rightarrow 0 = N^k v = \lambda^k v$. Over $\mathbb{C}$, conversely, if 0 is the only eigenvalue, the decomposition of Prop. 73 has the single summand $G(0, N)$, so $V = G(0, N)$ and $N$ is nilpotent.
- $N \neq 0$ is not diagonalizable: a diagonal matrix with all eigenvalues 0 is 0.

- Index $= \deg m_N \le n$. HW8 P1: $D$ on $\mathcal{P}_n(\mathbb{C})$ has index $n+1$, since $D^n x^n = n! \neq 0$ and $D^{n+1} = 0$.
- $(I-N)^{-1} = I + N + \cdots + N^{n-1}$: the product telescopes to $I - N^n = I$.
- Conversely, a strictly upper-triangular matrix is nilpotent: $N$ sends $\operatorname{span}(v_1, \ldots, v_j)$ into $\operatorname{span}(v_1, \ldots, v_{j-1})$, so $N^n = 0$. (Its only eigenvalue is 0 by Week 7, Prop. 60.) So $N$ is nilpotent $\iff$ $m_N = x^k$ for some $k$ $\iff$ $N$ has a strictly upper-triangular matrix in some basis (LADR 8.18 below) $\iff$ (over $\mathbb{C}$) 0 is the only eigenvalue.

**Proposition** (LADR 8.18: nilpotent operators are strictly upper-triangular)
If $N$ is nilpotent, then $V$ has a basis in which $\mathcal{M}(N)$ is upper-triangular with every diagonal entry 0.

**The idea**

We need a basis $v_1, \ldots, v_n$ with $Nv_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ for every $j$; that is exactly an upper-triangular matrix with zero diagonal.

If $N^k v = 0$, then $N^{k-1}(Nv) = 0$. So $N$ maps $\operatorname{null} N^k$ into $\operatorname{null} N^{k-1}$.

List vectors by the least power of $N$ that sends them to 0: a basis of $\operatorname{null} N$, extended to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$. A vector added at stage $k$ is sent into $\operatorname{null} N^{k-1}$, which is spanned by vectors listed earlier.

For example, $N = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ ($Ne_1 = e_2$, $Ne_2 = 0$): $\operatorname{null} N = \operatorname{span}(e_2)$, so list $e_2$ first. In the basis $e_2, e_1$ the matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

*Proof.* 1. Choose a basis of $\operatorname{null} N$, extend it to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$ (Props. 69, 70).

2. If $v$ was added for $\operatorname{null} N^k$, then $N^{k-1}(Nv) = N^k v = 0$, so $Nv \in \operatorname{null} N^{k-1}$, the span of the earlier layers (for $k = 1$, $Nv = 0$).

3. So each basis vector is sent into the span of the vectors listed before it: column $j$ has nonzero entries only above the diagonal.

□

**Theorem** (LADR 8.37: block diagonal form)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$ and $\mu_i = \dim G(\lambda_i, T)$. There is a basis in which

$$\mathcal{M}(T) = \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix},$$

where each $A_i$ is a $\mu_i \times \mu_i$ upper-triangular matrix with every diagonal entry equal to $\lambda_i$.

**The idea**

Prop. 73 splits $V$ into invariant pieces $G(\lambda_i, T)$, and invariant pieces give a block diagonal matrix (T9). So only the basis inside each piece is left to choose.

On $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with $N_i$ nilpotent, and $\lambda_i I$ has the matrix $\lambda_i I$ in every basis. So choose the basis that makes $N_i$ strictly upper-triangular (LADR 8.18); adding $\lambda_i I$ puts $\lambda_i$ on the diagonal.

*Proof.* 1. $N_i = (T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent, since $G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). By LADR 8.18, $G(\lambda_i, T)$ has a basis in which $\mathcal{M}(N_i)$ is upper-triangular with zero diagonal.

2. In that basis $\mathcal{M}(T|_{G(\lambda_i, T)}) = \lambda_i I + \mathcal{M}(N_i)$: upper-triangular, diagonal entries $\lambda_i$, size $\mu_i$.

3. $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73) with each piece invariant. The bases of the pieces put together form a basis of $V$ (T12), and in it the matrix is block diagonal (T9).

□

$$\mathcal{M}(T) = \left(\begin{array}{ccc|cc} \lambda_1 & * & * & 0 & 0 \\ 0 & \lambda_1 & * & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ \hline 0 & 0 & 0 & \lambda_2 & * \\ 0 & 0 & 0 & 0 & \lambda_2 \end{array}\right)$$

columns 1–3: a basis of $G(\lambda_1, T)$, $\mu_1 = 3$
columns 4–5: a basis of $G(\lambda_2, T)$, $\mu_2 = 2$

The zero blocks come from splitting $V$ into invariant pieces (Prop. 73, T9). Each diagonal block is $\lambda_i I$ plus a strictly upper-triangular nilpotent part (LADR 8.18).

**Proposition** (Chain lemma)
If $N^d v \neq 0$ and $N^{d+1} v = 0$, then $v, Nv, \ldots, N^d v$ are independent.

**The idea**

Applying $N^d$ to a term $c_j N^j v$ with $j \geq 1$ gives $c_j N^{j+d} v$, which is 0 because $j + d \geq d + 1$. Only the term $c_0 v$ survives, as $c_0 N^d v$. So $N^d$ isolates $c_0$; then $N^{d-1}$ isolates $c_1$, and so on. Same idea as Prop. 72: apply a map that sends every term but one to 0.

*Proof.* 1. Suppose $\sum_{j=0}^{d} c_j N^j v = 0$. We show by induction on $i$ that $c_0 = \cdots = c_i = 0$.

2. Assume $c_0 = \cdots = c_{i-1} = 0$, so the relation is $\sum_{j \geq i} c_j N^j v = 0$. Apply $N^{d-i}$: the term $c_j N^j v$ becomes $c_j N^{j+d-i} v$, which is 0 for $j > i$ because then $j + d - i \geq d + 1$.

3. What is left is $c_i N^d v = 0$, and $N^d v \neq 0$, so $c_i = 0$.

□

$$v \xrightarrow{N} Nv \xrightarrow{N} N^2 v \neq 0 \xrightarrow{N} 0$$

coefficient $c_0$ coefficient $c_1$ coefficient $c_2$

The case $d = 2$. Applying $N^2$ to $c_0 v + c_1 Nv + c_2 N^2 v$ sends each term two steps to the right. The $c_0$ term becomes $c_0 N^2 v$ with $N^2 v \neq 0$; the other two become 0. So $c_0 N^2 v = 0$ forces $c_0 = 0$.

**Theorem** (Prop. 75 / LADR 8.45: nilpotent normal form)
A nilpotent $N$ has a basis made of chains $u, Nu, \ldots, N^k u$ ($N^{k+1} u = 0$). Listing each chain as $N^k u, \ldots, Nu, u$, the matrix $\mathcal{M}(N)$ has only 1s and 0s directly above the diagonal and 0 everywhere else.

**The idea**

**The smallest case that shows everything.** On $\mathbb{F}^3$ let $Ne_3 = e_2$, $Ne_2 = 0$, $Ne_1 = 0$. A chain is a list $u, Nu, N^2 u, \ldots$ stopped at its last nonzero vector. Here $e_3, e_2$ is a chain of length 2 and $e_1$ is a chain of length 1, and together they are a basis. In the order $e_2, e_3, e_1$ the matrix of $N$ has a single 1 (from $Ne_3 = e_2$) and zeros elsewhere. The theorem says every nilpotent operator has a basis like this.

**Plan of the proof.** *Where to look for the chains.* Induction needs a smaller space that $N$ maps into itself. $U = \operatorname{range} N$ works: it is invariant, and it is smaller than $V$ because $N$ is not injective (T2). In the example $U = \operatorname{span}(e_2)$, with chain basis $e_2$.

*What a chain basis of $U$ is missing.* Compare it with what $V$ needs.

(1) Every chain of $U$ starts at a vector $v_i \in \operatorname{range} N$, so $v_i = Nu_i$ for some $u_i$. That $u_i$ is not in the list yet. Put it on top; the chain becomes one longer. (Example: $e_2 = Ne_3$, so add $e_3$.)

(2) A vector that $N$ sends to 0 but that is not in $\operatorname{range} N$ lies on no chain. (Example: $e_1$.) Such vectors must be added as chains of length 1.

*Which vectors of* $\operatorname{null} N$ *are already in the list?* Only the bottoms of the chains: the bottom is the one vector of a chain that $N$ sends to 0. So extend the bottoms to a basis of $\operatorname{null} N$ by new vectors $w_1, \ldots, w_l$.

*What is left to check.* That the new list has $n$ vectors, and that it is independent. The count is rank–nullity: the old list has $\dim \operatorname{range} N$ vectors, and we added one top per chain ($r$ of them) and $l$ vectors $w$, where $r + l = \dim \operatorname{null} N$.

Independence is the step that needs care. The plan: apply $N$ to a relation. That removes the $w$'s and the bottoms and turns everything else into vectors of the old chain basis of $U$, which we know is independent.

*Proof.* 1. If $N = 0$, any basis works (chains of length 1). Otherwise $U = \operatorname{range} N$ is invariant with $0 < \dim U < n$ ($U \neq 0$ because $N \neq 0$; $\dim U < n$ because $N$ is not injective, T2), and $N|_U$ is nilpotent since $N^k = 0$ on $V$. By induction $U$ has a chain basis $v_i, Nv_i, \dots, N^{d_i}v_i$ for $i = 1, \dots, r$, with $N^{d_i+1}v_i = 0$. Write $b_i = N^{d_i}v_i$ for the bottoms.

2. *The bottoms are a basis of* $\operatorname{null} N \cap U$. They lie in $\operatorname{null} N$ and in $U$, and they are independent (part of a basis). They span: take $x \in U$ with $Nx = 0$ and write $x = \sum_{i,j} c_{ij} N^j v_i$. Then $0 = Nx = \sum_{j<d_i} c_{ij} N^{j+1} v_i$ (the terms with $j = d_i$ are 0). The vectors $N^{j+1}v_i$ with $j < d_i$ are different members of the chain basis, so $c_{ij} = 0$ for $j < d_i$, and $x = \sum_i c_{id_i} b_i$.

3. Choose $u_i$ with $Nu_i = v_i$ (possible since $v_i \in \operatorname{range} N$). Extend $b_1, \dots, b_r$ by $w_1, \dots, w_l$ to a basis of $\operatorname{null} N$. So $\dim \operatorname{null} N = r + l$.

4. *The new list* is $u_i, v_i, Nv_i, \dots, b_i$ for each $i$, together with $w_1, \dots, w_l$. *Count,* using rank–nullity (T1) at the last step:
$$\sum_i (d_i + 2) + l = \sum_i (d_i + 1) + r + l = \dim \operatorname{range} N + \dim \operatorname{null} N = n.$$

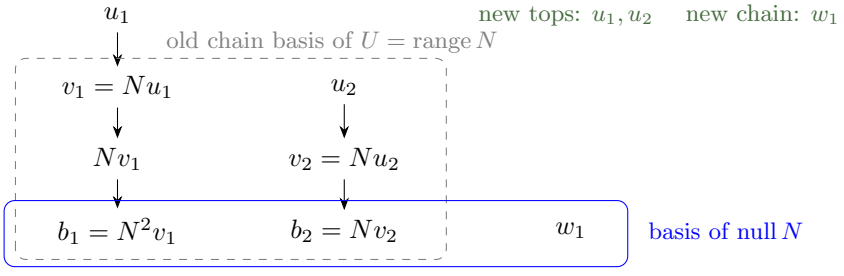
5. *Independence, first half.* Suppose $\sum_i a_i u_i + \sum_{i,j} c_{ij} N^j v_i + \sum_k e_k w_k = 0$. Apply $N$. We have $Nu_i = v_i$; $N(N^j v_i) = N^{j+1} v_i$, which is 0 for $j = d_i$; and $Nw_k = 0$. So
$$\sum_i a_i v_i + \sum_{j<d_i} c_{ij} N^{j+1} v_i = 0.$$
The vectors here are different members of the chain basis of $U$ (the tops $v_i$, and the vectors below them). So every $a_i = 0$, and $c_{ij} = 0$ for $j < d_i$.

6. *Independence, second half.* What is left of the relation is $\sum_i c_{id_i} b_i + \sum_k e_k w_k = 0$. The $b_i$ and $w_k$ together are a basis of $\operatorname{null} N$ (step 3), so these coefficients are 0 too. This is the reason the $w$'s were chosen to extend the bottoms: a $w$ that was a combination of chain vectors would, by step 5, be a combination of the bottoms, and then bottoms and $w$'s would be dependent.

7. So the new list is $n$ independent vectors, a basis (T3), and it consists of chains. Listing a chain as $N^k u, \dots, Nu, u$, $N$ sends each vector to the one before it and the first to 0. That is a column with a single 1 directly above the diagonal, or a zero column.

□

For example, the two matrices below are in this form; the first has chains of lengths $2, 2$, the second one chain of length 4:
$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$




Two chains of $U$ (inside the dashed box) with $d_1 = 2$, $d_2 = 1$. Each gets a new top $u_i$ with $Nu_i = v_i$. The bottoms $b_1, b_2$ are extended by $w_1$ to a basis of $\operatorname{null} N$ (bottom row). Arrows are $N$; the bottom row is sent to 0.

**Definition** (Def. 40)**. Jordan block**: $J_k(\lambda) = \lambda I + N_k$ ($k \times k$, 1s on the superdiagonal), so $Je_1 = \lambda e_1$, $Je_j = \lambda e_j + e_{j-1}$. **Jordan normal form**: block diagonal with Jordan blocks. **Jordan basis**: a basis giving Jordan form.

The $\lambda$'s of different blocks may be equal. For example, the matrices below have 2, 3 and 5 Jordan blocks:

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}.$$

**Example (HW8 P1): differentiation is one Jordan block.** $D$ on $\mathcal{P}_n(\mathbb{C})$ is the single block $J_{n+1}(0)$ in the basis $1, x, x^2/2!, \ldots, x^n/n!$, because $D$ sends each basis vector to the previous one:

$$D\frac{x^k}{k!} = \frac{x^{k-1}}{(k-1)!}.$$

**Theorem** (Prop. 76 / LADR 8.46: Jordan normal form)

Every operator on a finite-dimensional complex space has a Jordan basis.

**The idea**

Prop. 73 splits $V$ into invariant pieces $G(\lambda_i, T)$, so the matrix is block diagonal (T9). On each piece $T = \lambda_i I + N_i$ with $N_i$ nilpotent, and $\lambda_i I$ has the same matrix in every basis.

Use the chain basis of $N_i$ (Prop. 75) in each piece: $N_i$ becomes a sum of blocks $N_k$, so $T$ becomes a sum of blocks $\lambda_i I + N_k = J_k(\lambda_i)$. This is the proof of LADR 8.37 with the chain basis in place of the triangular one.

*Proof.* 1. $V = \bigoplus_i G(\lambda_i, T)$ with each piece invariant (Prop. 73, Prop. 70).

2. $N_i = (T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent (Prop. 70), so by Prop. 75 the piece has a basis of chains in which $\mathcal{M}(N_i)$ is block diagonal with blocks $N_k$.

3. In that basis $\mathcal{M}(T|_{G(\lambda_i,T)}) = \lambda_i I + \mathcal{M}(N_i)$, block diagonal with blocks $\lambda_i I + N_k = J_k(\lambda_i)$.

4. The bases of the pieces put together form a basis of $V$ (T12), and in it the matrix of $T$ is block diagonal (T9) with Jordan blocks.

□

**Compared with LADR 8.37.** Both start by splitting $V$ into the $G(\lambda_i, T)$. LADR 8.37 only makes each block upper-triangular with $\lambda_i$ on the diagonal; Jordan form also makes every entry above the diagonal 0, except the entries directly above it, which are 0 or 1. That needs the chain construction of Prop. 75.

**Reading the Jordan form.** For each eigenvalue $\lambda$:

- number of blocks for $\lambda$ = geometric multiplicity $\dim E(\lambda, T)$;
- total size of those blocks = algebraic multiplicity $\dim G(\lambda, T)$;
- largest block for $\lambda$ = exponent of $(x-\lambda)$ in $m_T$;
- $\chi_T = \prod (x-\lambda)^{\text{alg. mult.}}$, $m_T = \prod (x-\lambda)^{\text{largest block}}$;
- diagonalizable $\iff$ all blocks $1\times 1$ $\iff$ $m_T$ has no repeated root.

Instance: $J_3(4) \oplus J_2(4) \oplus J_1(-1)$ has $\chi = (x-4)^5(x+1)$, $m = (x-4)^3(x+1)$, and $\dim E(4) = 2$.

**Why.**

1. One block $J_k(\lambda)$: $J_k(\lambda) - \lambda I = N_k$ moves each basis vector one place ($e_j \mapsto e_{j-1}$, $e_1 \mapsto 0$), so $N_k^j$ moves it $j$ places and $\dim \operatorname{null} N_k^j = \min(j,k)$. Hence $E = \operatorname{span}(e_1)$ (one block, dimension 1), $G$ is the whole block (size $k$), and $m = \chi = (x-\lambda)^k$.
2. A block $J_j(\mu)$ with $\mu \neq \lambda$ adds nothing to $\operatorname{null}(T-\lambda I)^k$: $(J_j(\mu) - \lambda I) = (\mu-\lambda)I + N_j$ is invertible (a nilpotent has only the eigenvalue 0, so $N_j v = (\lambda-\mu)v$ forces $v = 0$), and so are its powers.
3. For the whole matrix, null spaces add over the blocks (T9) and $m$ is the lcm of the blocks' minimal polynomials (T11(e)). Adding up gives each line of the box.

**Theorem** (Prop. 77: counting blocks)

Over $\mathbb{C}$, let $d_k = \dim \operatorname{null}(T-\lambda I)^k$ ($d_0 = 0$). For $k \geq 1$:

- number of blocks for $\lambda$ of size $\geq k = d_k - d_{k-1}$;
- number of size exactly $k = 2d_k - d_{k-1} - d_{k+1}$.

So the Jordan form is unique up to order, and two complex matrices are similar iff they have the same Jordan form.

**The idea**

The numbers $d_k$ are defined from $T$ alone, without choosing a basis; so it is enough to express the block sizes through them. Compute $d_k$ in a Jordan basis. Null spaces split block by block (T9), so $d_k$ is a sum over the blocks. For one block, $J_j(\lambda) - \lambda I$ is a shift, and the null space of its $k$-th power has dimension $\min(k,j)$: it rises by 1 per step until $k = j$, then stays. So from $k-1$ to $k$, exactly the blocks of size $\geq k$ add 1 (blocks for other eigenvalues add 0).

**Example: counting blocks.** For the blocks $J_3(0), J_1(0)$, $d_1, d_2, d_3 = 2, 3, 4$. So there are 2 blocks of size $\geq 1$, 1 of size $\geq 2$, and 1 of size $\geq 3$.

*Proof.* 1. In a Jordan basis, $d_k = \sum_{\text{blocks}} \dim \operatorname{null}(J-\lambda I)^k$. Blocks for $\mu \neq \lambda$ contribute 0; $J_j(\lambda)$ contributes $\min(k,j)$.

2. $\min(k,j) - \min(k-1,j)$ is 1 if $j \geq k$, else 0. Summing:

$$d_k - d_{k-1} = \#\{\text{blocks with } j \geq k\}.$$

Exactly $k$: $(d_k - d_{k-1}) - (d_{k+1} - d_k)$.

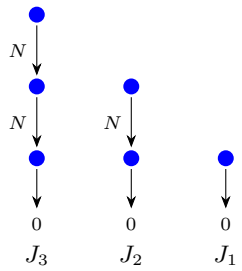
3. Similar matrices are one operator in two bases (T10), so they have the same $d_k$ and the same Jordan form. Conversely, matrices with the same Jordan form are both similar to it.

□

**The question answered: $I$, $B$, $C$.** The three matrices below have ranks 1, 2, 1, and $(B-I)^2 = 0$, $(C-I)^3 = 0$:

$$B - I = \begin{pmatrix} 2 & -1 & -1 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{pmatrix}, \qquad C - I = \begin{pmatrix} 2 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 2 & -2 \end{pmatrix}, \qquad (C-I)^2 = \begin{pmatrix} 2 & -4 & 2 \\ 2 & -4 & 2 \\ 2 & -4 & 2 \end{pmatrix}.$$

1. For $B$: $d_1, d_2 = 2, 3$, so there are 2 blocks, 1 of size $\geq 2$: the Jordan form is $J_2(1) \oplus J_1(1)$.
2. For $C$: $d_1, d_2, d_3 = 1, 2, 3$: one block, $J_3(1)$.
3. $I$ is $J_1(1) \oplus J_1(1) \oplus J_1(1)$. The three Jordan forms differ, so $I$, $B$, $C$ are pairwise not similar (Prop. 77).




row 3: 1 dot, $d_3 = 5 + 1 = 6$

row 2: 2 dots, $d_2 = 3 + 2 = 5$

row 1: 3 dots, $d_1 = 3$

A nilpotent $N$ with blocks $J_3(0), J_2(0), J_1(0)$. Each column is one chain, read downward; the bottom dot is an eigenvector. $N^k$ sends exactly the dots in the bottom $k$ rows to 0, so $d_k$ is the number of dots in rows $1, \ldots, k$, and $d_k - d_{k-1}$ is the number of dots in row $k$, i.e. the number of blocks of size $\geq k$.

**Example 63: Jordan form from null-space dimensions.** $T \in \mathcal{L}(\mathbb{C}^8)$ with eigenvalues $1, 2$. Method: list $d_0, d_1, \ldots$, take differences, then differences of differences.

| $\lambda$ | $d_0$ | $d_1$ | $d_2$ | $d_3$ | $d_4$ | blocks |
|---|---|---|---|---|---|---|
| 1 | 0 | 2 | 4 | 4 | | sizes $\geq 1$: 2, $\geq 2$: 2, $\geq 3$: 0 $\Rightarrow J_2(1), J_2(1)$ |
| 2 | 0 | 2 | 3 | 4 | 4 | sizes $\geq 1$: 2, $\geq 2$: 1, $\geq 3$: 1, $\geq 4$: 0 $\Rightarrow J_3(2), J_1(2)$ |

($d_4 = 4$ for $\lambda = 2$ because $\dim G(2, T) = 8 - 4$.) Answer: $J_2(1) \oplus J_2(1) \oplus J_3(2) \oplus J_1(2)$. The formula for size exactly $k$ gives the same: for $\lambda = 2$, $k = 3$, $2 \cdot 4 - 3 - 4 = 1$ block of size 3; $k = 1$, $2 \cdot 2 - 0 - 3 = 1$ block of size 1.

**Example 62: finding a Jordan basis.** The matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

has only the eigenvalue 1, and $d_1, d_2, d_3 = 1, 2, 3$, so it has one block $J_3(1)$.

1. Pick the top: $v_3 \in \operatorname{null}(A-I)^3$, $v_3 \notin \operatorname{null}(A-I)^2$. Here $v_3 = e_3$.
2. Go down: $v_2 = (A-I)v_3 = (3, 2, 0)$, $v_1 = (A-I)v_2 = (4, 0, 0)$ (an eigenvector).
3. Check: $Av_1 = v_1$, $Av_2 = v_2 + v_1$, $Av_3 = v_3 + v_2$. That is $J_3(1)$.
4. So with $S$ the matrix whose columns are $v_1, v_2, v_3$, $A = SJ_3(1)S^{-1}$ (T10):

$$S = \begin{pmatrix} 4 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$v_3 = (0,0,1) \xrightarrow{A-I} v_2 = (3,2,0) \xrightarrow{A-I} v_1 = (4,0,0) \xrightarrow{A-I} 0$$

top: not in $\operatorname{null}(A-I)^2$ — eigenvector

Start at the top and apply $A-I$ until the next step gives 0. In the basis $(v_1, v_2, v_3)$, read from the bottom of the chain, each vector is sent to the one before it, which puts the 1s just above the diagonal.

Why start at the top: the top vector determines the whole chain, and a top outside $\operatorname{null}(A-\lambda I)^{k-1}$ guarantees full length $k$ (chain lemma).

**Several blocks.** Get the block sizes first (Prop. 77). For each block of size $j$, largest first, pick a top $v \in \operatorname{null}(A-\lambda I)^j \setminus \operatorname{null}(A-\lambda I)^{j-1}$ and keep its chain only if its bottom $(A-\lambda I)^{j-1}v$ is independent of the bottoms already chosen. (Independent of the vectors so far is not enough: in $J_3(0) \oplus J_2(0)$ with chains $e_1, e_2, e_3$ and $f_1, f_2$, the top $f_1 + e_2$ gives a chain ending at $e_1$ again.)

**Proposition** (Chains with independent bottoms are independent)

Let $N$ be nilpotent and $u_1, \dots, u_r$ vectors with $N^{k_i}u_i = 0$, $N^{k_i-1}u_i \neq 0$, such that the bottoms $N^{k_1-1}u_1, \dots, N^{k_r-1}u_r$ are independent. Then all the vectors $N^j u_i$, $0 \le j < k_i$, are independent.

*Proof.* 1. Induction on the total number $L = k_1 + \cdots + k_r$ of vectors. If every $k_i = 1$, the vectors are the bottoms, which are independent.

2. Suppose $\sum_{i,j} c_{ij} N^j u_i = 0$ and apply $N$. The terms with $j = k_i - 1$ become 0, since $N^{k_i}u_i = 0$; the others become $c_{ij}N^{j+1}u_i$.

3. The vectors $N^{j+1}u_i$ with $j < k_i - 1$ are the chains of the vectors $Nu_i$ (for the $i$ with $k_i \ge 2$), of lengths $k_i - 1$. Their bottoms are the same $N^{k_i-1}u_i$, some of the original bottoms, so they are independent. These chains have fewer than $L$ vectors in total, so by induction $c_{ij} = 0$ for all $j < k_i - 1$.

4. What is left of the relation is $\sum_i c_{i,k_i-1} N^{k_i-1}u_i = 0$, a relation among the bottoms, so these coefficients are 0 too.

□

So the procedure gives an independent list of the right length, a Jordan basis (T3).

**Powers.** $J = \lambda I + N$ with $\lambda I$ and $N$ commuting, so the binomial theorem applies and $N^j = 0$ for $j \ge k$:

$$J_k(\lambda)^m = \sum_{j=0}^{\min(m,k-1)} \binom{m}{j} \lambda^{m-j} N^j, \qquad J_3(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} & \binom{m}{2}\lambda^{m-2} \\ 0 & \lambda^m & m\lambda^{m-1} \\ 0 & 0 & \lambda^m \end{pmatrix}.$$

$J_3(1)^{100}$. $J = I + N$ with $N^3 = 0$, so $J^{100} = I + 100N + \binom{100}{2}N^2$, and $\binom{100}{2} = 4950$:

$$J_3(1)^{100} = \begin{pmatrix} 1 & 100 & 4950 \\ 0 & 1 & 100 \\ 0 & 0 & 1 \end{pmatrix}.$$

- Any Jordan matrix is $J = D + N$, $D$ diagonal, $N$ nilpotent, $DN = ND$: on a block, $D$ is $\lambda I$, which commutes with everything, and the products $DN$, $ND$ are computed block by block. This is the Jordan–Chevalley decomposition. If $J$ is invertible, also $J = DU$ with $U = I + D^{-1}N$, $DU = UD$.
- **Square roots (LADR 8.39–8.41).** $I+N$ has a square root: choose $a_j$ with $(I + a_1N + \cdots + a_{n-1}N^{n-1})^2 = I+N$ by matching coefficients: the coefficient of $N^j$ in the square is $2a_j + \sum_{i=1}^{j-1} a_i a_{j-i}$, which must be 1 for $j = 1$ and 0 for $j \ge 2$, so each $a_j$ is determined by the earlier ones ($a_1 = \frac{1}{2}$, $a_2 = -\frac{1}{8}$, …); powers $N^n$ and up vanish. This is the series
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$$
cut off. Over $\mathbb{C}$, every invertible operator has a square root: on each $G(\lambda_i, T)$, $T = \lambda_i(I + N_i/\lambda_i)$ with $\lambda_i \neq 0$. Take a complex number $\nu_i$ with $\nu_i^2 = \lambda_i$ and a square root $R_i$ of $I + N_i/\lambda_i$ as above; then $(\nu_i R_i)^2 = T$ on $G(\lambda_i, T)$. Put these together on $V = \bigoplus_i G(\lambda_i, T)$.

**The question answered.** Yes: the Jordan form (Prop. 76), and two complex matrices are similar exactly when their Jordan forms agree up to order (Prop. 77). For $B$ it is $J_2(1) \oplus J_1(1)$ and for $C$ it is $J_3(1)$ (the example after Prop. 77).

**Shortcuts**

- **Block counts from ranks.** The number of blocks for $\lambda$ of size $\geq k$ is $\operatorname{rank}(A-\lambda I)^{k-1} - \operatorname{rank}(A-\lambda I)^k$. Reason: Prop. 77 and (T1). Instance: Ex. 62 has ranks $3,2,1,0$, so there is one block of size $\geq 1$, $\geq 2$ and $\geq 3$: $J_3(1)$.
- **The increments $d_k - d_{k-1}$ never increase,** since they count blocks of size $\geq k$. Reason: Prop. 77. Instance: the data $d_1, d_2, d_3 = 2, 3, 5$ are impossible (increments $2, 1, 2$).
- **Small multiplicities.** If $\lambda$ has algebraic multiplicity $\leq 3$, the exponent of $x - \lambda$ in $m_T$ fixes the blocks: multiplicity 3 with exponent $1, 2, 3$ gives sizes 1+1+1, 2+1, 3. Reason: the sizes add up to the multiplicity and the largest is the exponent. Instance: for the matrix $C$ of HW7 P6 (the matrix of the Exponent 1 test in Section 2.1), 2 has multiplicity 2 and exponent 2, so $J_2(2) \oplus J_1(3)$. For multiplicity 4 this fails (first trap).
- **Powers of a matrix.** If $A = SJS^{-1}$, then $A^m = SJ^mS^{-1}$, and $J^m$ is computed block by block with the binomial formula. Reason: the inner factors $S^{-1}S$ cancel. Instance: for Ex. 62, $A^m = S\,J_3(1)^m S^{-1}$.
- **Inverse of $\lambda I + N$, $\lambda \neq 0$, $N$ nilpotent:** $(\lambda I + N)^{-1} = \sum_{j=0}^{n-1} (-1)^j \lambda^{-j-1} N^j$. Reason: multiplying out, the sum telescopes to $I - (-N/\lambda)^n = I$. Instance: $J_2(2)^{-1} = \left[\begin{smallmatrix} 1/2 & -1/4 \\ 0 & 1/2 \end{smallmatrix}\right]$.
- $N^2 = 0$ **forces** $\operatorname{rank} N \leq n/2$**.** Reason: $\operatorname{range} N \subseteq \operatorname{null} N$, so $\operatorname{rank} N \leq n - \operatorname{rank} N$ (T1). Instance: a $3 \times 3$ $N$ with $N^2 = 0$ has rank $\leq 1$, so it is 0 or has the blocks $J_2(0) \oplus J_1(0)$.

**Traps**

- $\chi_T$ and $m_T$ do not fix the Jordan form: $J_2(0) \oplus J_2(0)$ and $J_2(0) \oplus J_1(0) \oplus J_1(0)$ both have $\chi = x^4$, $m = x^2$.
- Adding the number of blocks is still not enough: $J_3 \oplus J_3 \oplus J_1$ and $J_3 \oplus J_2 \oplus J_2$ (all for 0) agree in $\chi$, $m$ and block count, but $d_2 = 5$ vs. 6. You need all the $d_k$.
- Order each chain from the bottom (the eigenvector) up; reversed, the 1s go below the diagonal: $J_2(0)$ in the order $e_2, e_1$ has the matrix $\left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$.
- Jordan form needs $\mathbb{C}$ (or split $m_T$): the rotation of $\mathbb{R}^2$ has none over $\mathbb{R}$.
- The exponent of $\lambda$ in $m_T$ is the size of the largest block for $\lambda$, not the number of blocks (that is $\dim E(\lambda, T)$): $J_3(0)$ has one block but $m = x^3$.

## §3 Problem set

**HW8 P1.** Let $D\colon \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$ be differentiation.

(a) Prove that $D$ is nilpotent and determine its nilpotency index.

(b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.

(c) Find a basis with respect to which the matrix of $D$ is a single Jordan block.

*Hint:* $D^{n+1} = 0$ but $D^n x^n = n!$; $E(0, D)$ is the constants; for (c) the basis $x^k/k!$ (the example after Def. 40).

**HW8 P2.** Construct explicit matrices having the following spectral data.

(a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue $-1$ has algebraic multiplicity 2 and geometric multiplicity 2.

(b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

*Hint:* "Reading the Jordan form": number of blocks = geometric multiplicity, total size = algebraic multiplicity, largest block = exponent in $m$.

**HW8 P3.** Let

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Compute $\operatorname{null}(J - I)$, $\operatorname{null}((J - I)^2)$, and $\operatorname{null}((J - I)^3)$.

(b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.

(c) Compute $J^m$ for every positive integer $m$.

*Hint:* $J - I$ is the shift $e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$; for (c) use the binomial theorem with $N^3 = 0$ (Powers).

**HW8 P4.** Let $N$ be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

(a) Compute $\dim \operatorname{null} N^k$ for $k = 0, 1, 2, 3, 4$.

(b) Suppose instead that you are only told $\dim \operatorname{null} N = 2$, $\dim \operatorname{null} N^2 = 4$, $\dim \operatorname{null} N^3 = 5$. Explain how these numbers force the same two block sizes.

*Hint:* $\dim \operatorname{null} J_j(0)^k = \min(k, j)$, added over the blocks (Prop. 77); in (b) take differences and use $\dim V = 5$.

**HW8 P5. (optional)** Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \operatorname{null}(T - I) = 1, \quad \dim \operatorname{null}((T - I)^2) = 2, \quad \dim \operatorname{null}((T - I)^3) = 2,$$

while

$$\dim \operatorname{null}(T - 2I) = 2, \quad \dim \operatorname{null}((T - 2I)^2) = 3, \quad \dim \operatorname{null}((T - 2I)^3) = 4, \quad \dim \operatorname{null}((T - 2I)^4) = 4.$$

Determine the Jordan normal form of $T$, up to reordering of the blocks.

*Hint:* Prop. 77: the table of $d_k$ as in Ex. 63.

**HW8 P6. (optional)** Suppose an operator has Jordan blocks $J_3(4)$, $J_2(4)$, $J_1(-1)$. Determine:

(a) its characteristic polynomial;

(b) its minimal polynomial;

(c) the geometric and algebraic multiplicities of each eigenvalue;

(d) whether it is diagonalizable.

*Hint:* "Reading the Jordan form".

**HW8 P7. (optional)** Prove that for a nilpotent operator $N$, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ is the number of Jordan blocks of size at least $k$.

*Hint:* Prop. 77, steps 1–2: compute in a Jordan basis, one block at a time.

**G1.** Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

act on $\mathbb{C}^3$. Find the eigenvalues, $E(2, A)$, $G(2, A)$, the Jordan form, a Jordan basis, and $m_A$.

*Hint:* the $2 \times 2$ block has trace 4 and determinant 4; then $\operatorname{rank}(A - 2I)$ and $(A - 2I)^2$; build the chain from a top outside $\operatorname{null}(A - 2I)$ (Ex. 62).

**G2.** Let $V$ be finite-dimensional and $S, T \in \mathcal{L}(V)$ with $ST = TS$. Prove that $G(\lambda, T)$ is invariant under $S$.

*Hint:* $S$ commutes with $(T - \lambda I)^n$, and $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$ (Prop. 70).

**G3.**

(a) Let $N$ be nilpotent on a finite-dimensional $V \neq \{0\}$. Prove that $\lambda I + N$ is invertible if and only if $\lambda \neq 0$.

(b) Let $V$ be a finite-dimensional complex vector space and suppose $T \in \mathcal{L}(V)$ has only one eigenvalue $\lambda$. Prove that $T - \lambda I$ is nilpotent.

*Hint:* (a) the only eigenvalue of $N$ is 0 (bullets after Def. 39); (b) Prop. 73 with a single eigenvalue.

**G4.** Find all possible Jordan forms of $T \in \mathcal{L}(\mathbb{C}^4)$ with $\chi_T = (x-2)^4$ and $m_T = (x-2)^2$. Which single number tells them apart?

*Hint:* the block sizes add up to 4 and the largest is 2.

**G5.** Let $V$ be a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that if every generalized eigenvector of $T$ is an eigenvector, then $T$ is diagonalizable.

*Hint:* show $G(\lambda, T) \subseteq E(\lambda, T)$, then Prop. 73 and the criterion after Def. 38.

**R1 (Midterm 2 review, Problem 5).** Find the minimal polynomial, the characteristic polynomial, the eigenvalues and the eigenspaces of

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

as an operator on $\mathbb{R}^3$.

*Hint:* triangular: LADR 8.31 gives $\chi = (x-3)^2(x-1)$; test $(x-3)(x-1)$.

**R2 (Midterm 2 review, Problem 8: true or false).** The dimension of $E(\lambda, T)$ is the multiplicity of $x - \lambda$ in the minimal polynomial.

*Hint:* "Reading the Jordan form".